

SARVESHKUMAR KUMARAGURUPARAN

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Education

University of Southern California [USC]

Master of Science in Computer Science

Aug. 2025 – Present

Los Angeles, California

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science

Sep. 2021 – May 2025

Chennai, India

Relevant Coursework

- Machine Learning
- Deep Learning
- Advanced Python
- Artificial Intelligence
- Multivariate Data Analysis
- Database Management

Experience

Mayaa App

GenAI Intern

Apr. 2025 – Aug. 2025

Mumbai, India

- Built Llama 4 Scout-based system to convert unstructured WhatsApp customer data into JSON format, replacing legacy regex processing and improving data accuracy.
- Deployed complete AWS pipeline with API integrations and cost optimization, delivering scalable solution with enhanced operational efficiency.

TAIIN Steel Fab And Infra Private Limited

Machine Learning Intern

Sep. 2024 – Mar. 2025

Remote, India

- Developed neural network models for product demand forecasting and implemented custom optimization algorithms for transportation logistics, improving supply chain efficiency and reducing inventory waste.
- Validated model performance across product categories and delivered scalability recommendations to operations teams, enabling company-wide predictive analytics adoption.

National University of Singapore [NUS]

Deep Learning Research Intern

Jun. 2024 – Jul. 2024

Singapore

- Developed custom YOLO model for human pose recognition, achieving 87% accuracy - the highest performance recorded for this specific model architecture.
- Successfully deployed the optimized pose recognition system on AWS infrastructure for production-level implementation.

Projects

Marine Microplastic Detection using Satellite data | *Python, SNAP, Copernicus*

Feb. 2025

- Integrated Sentinel-1 SAR imagery with CYGNSS data and enhanced Random Forest classifier using dynamic thresholds, achieving 91.30% detection accuracy and improving F1 score from 86.75% to 95.44%.
- Validated framework against 114 in-situ measurements, demonstrating scalable system capabilities for marine conservation applications.

Adaptive Real-Time Human Pose Estimation for Targeted Physical Therapy | *Python, Keypoints*

Sep 2024

- Designed custom CNN-RNN model with cross-attention for real-time shoulder rehabilitation, providing adaptive feedback to guide patients in maintaining proper posture during unsupervised physical therapy.
- Conducted comprehensive system testing and optimization to ensure accuracy and usability for real-time pose estimation and patient guidance.

Demographic Determinants of ADHD: Analyzing Prevalence and Diagnosis Patterns | *Python*

Nov. 2023

- Analyzed ADHD prevalence patterns across demographic factors (age, gender, socioeconomic status) using Random Forest and Neural Networks, revealing higher male diagnosis rates and age-related symptom variations.
- Developed predictive models with Scikit-Learn and TensorFlow to enable targeted interventions and improve diagnostic accuracy based on demographic trends.

Technical Skills

Languages: Python, C++, SQL, R

ML/DL Frameworks: TensorFlow, PyTorch, Scikit-Learn, Keras, XGBoost, OpenCV

Cloud & MLOps: AWS (SageMaker, EC2, S3, Lambda), Docker

Specialized: Computer Vision, NLP, A/B Testing

Extracurricular

Study Abroad Program

Advanced Statistics, Machine Learning Theory

Fall 2023 – Present

UC Davis